Transformations

Name _____ Date _____ No calculator Version 1

Scan to practise this topic online

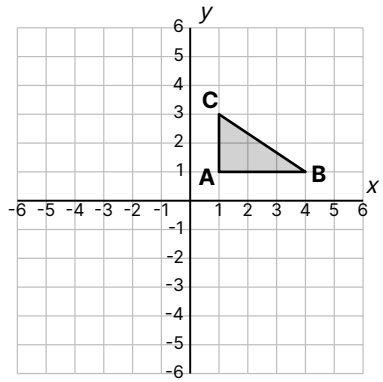
KEY WORDS object *(ahanoa)* · image *(ataata)* · translation *(nekehanga)* · reflection *(whakaata)*

A Translation ●○○

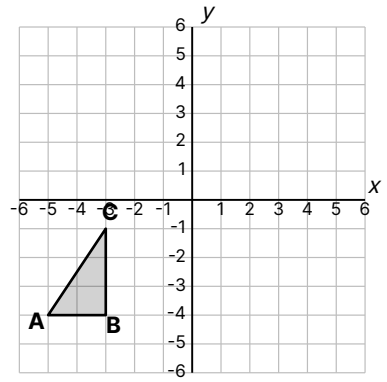
Remember: A translation slides every point the same distance in the same direction. Label the image A′, B′, C′.

1. Draw the image of each shape. Then write the coordinates of its corners.

a) Translate the shape 5 units left and 2 units up.



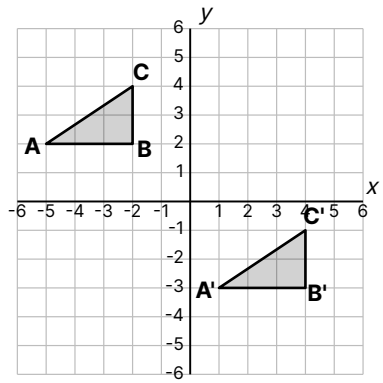
b) Translate the shape 7 units right and 3 units up.



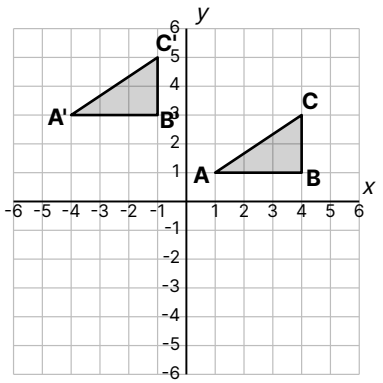
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2. Each shape ABC has been translated to A′B′C′. Describe the translation.

a)



b)

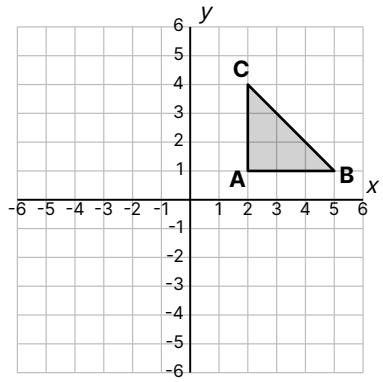


B Reflection ●●○

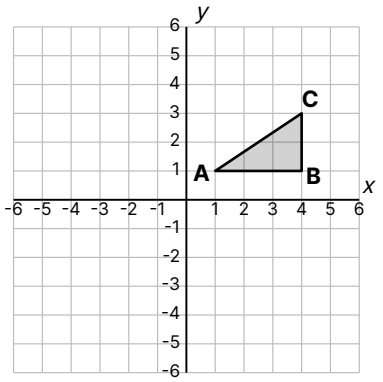
Remember: Each point and its image are the same distance from the mirror line, on opposite sides of it.

3. Draw the image of each shape. Then write the coordinates of its corners.

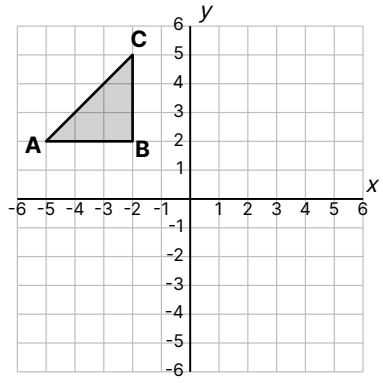
a) Reflect the shape in the y-axis.



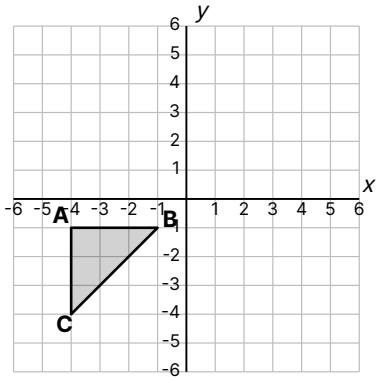
b) Reflect the shape in the x-axis.



c) Reflect the shape in the y-axis.



d) Reflect the shape in the x-axis.



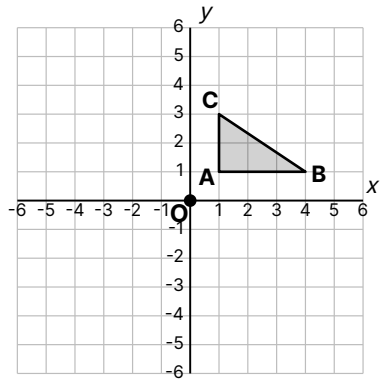
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C Rotation ●●●

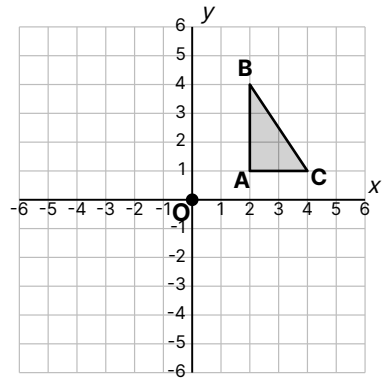
Remember: Every point turns through the same angle about the centre. Tracing paper helps: trace the shape, hold the centre still and turn.

4. Draw the image of each shape. Then write the coordinates of its corners.

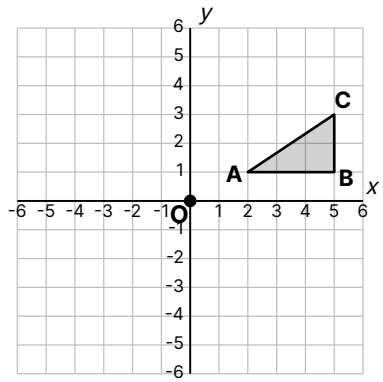
a) Rotate the shape 180° about the point O.



b) Rotate the shape 90° clockwise about the point O.

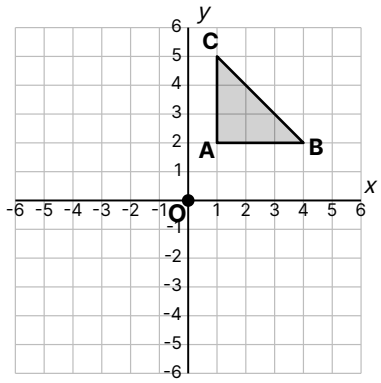


c) Rotate the shape 90° anticlockwise about the point O.



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d) Rotate the shape 90° clockwise about the point O.

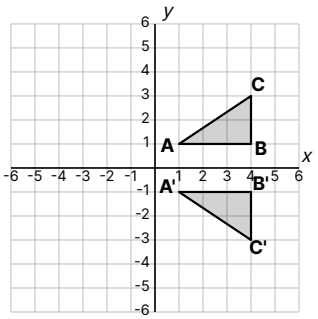


D Naming the transformation ●●●

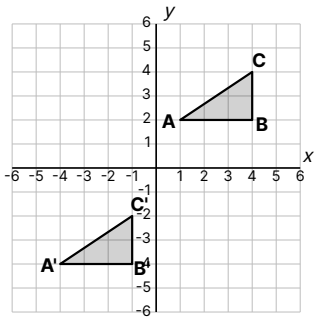
Remember: Translation: a slide. Reflection: a flip. Rotation: a turn.

5. Shape ABC has been transformed to A′B′C′. Name and describe each transformation.

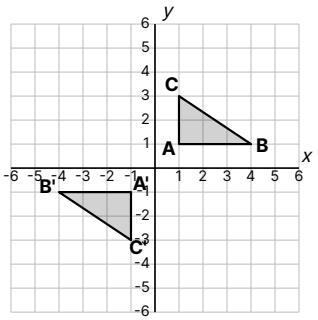
a)



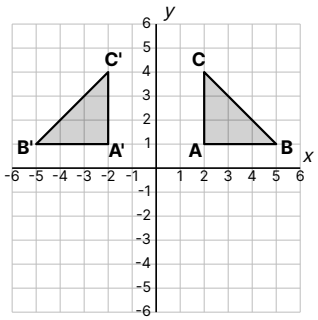
b)



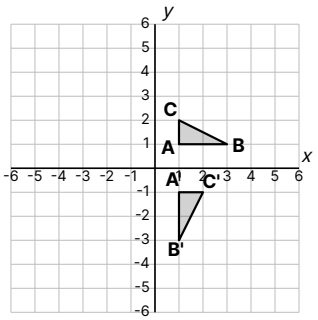
c)



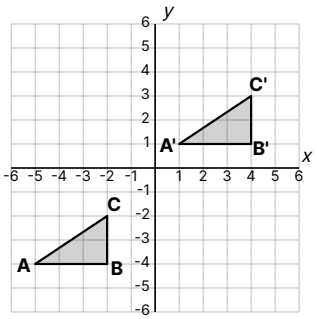
d)



e)



f)



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Extension

6. Without drawing, write the coordinates of the image of the point $P(4,\ 2)$ after each transformation.

a) Translation 6 left and 1 up _____

b) Reflection in the x-axis _____

c) Reflection in the y-axis _____

d) Rotation of 180° about (0, 0) _____

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Transformations

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

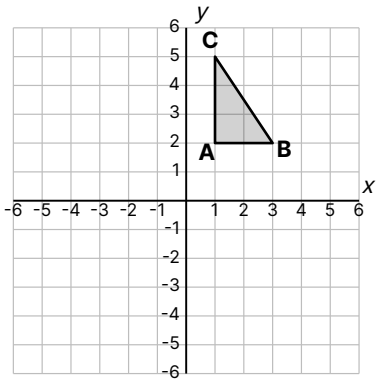
KEY WORDS object *(ahanoa)* · image *(ataata)* · translation *(nekehanga)* · reflection *(whakaata)*

A Translation ●○○

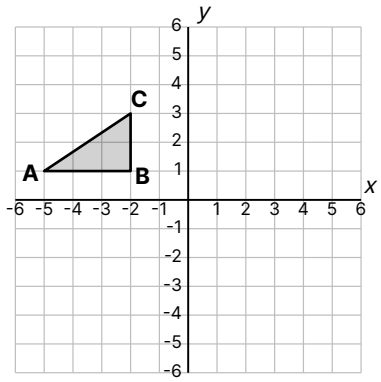
Remember: A translation slides every point the same distance in the same direction. Label the image A′, B′, C′.

1. Draw the image of each shape. Then write the coordinates of its corners.

a) Translate the shape 6 units left and 4 units down.



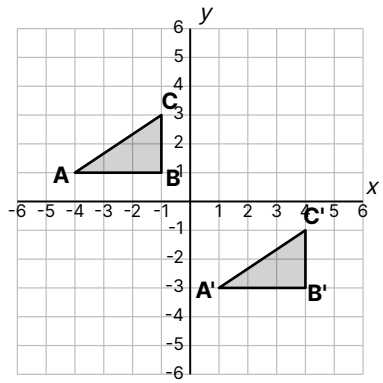
b) Translate the shape 6 units right and 5 units down.



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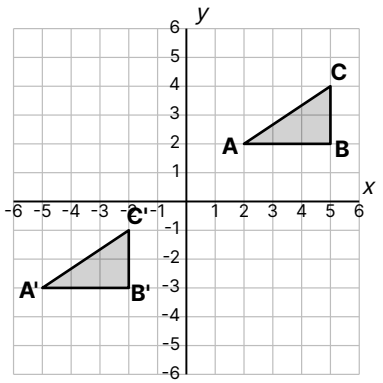
2. Each shape ABC has been translated to A′B′C′. Describe the translation.

a)



_____

b)



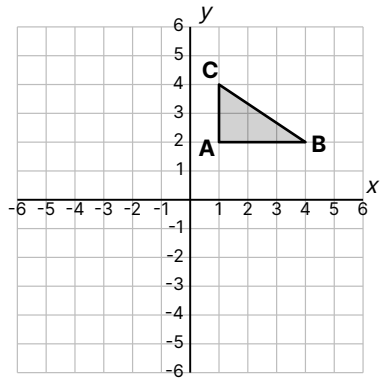
_____

B Reflection ●●○

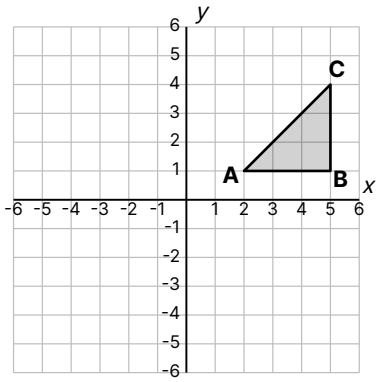
Remember: Each point and its image are the same distance from the mirror line, on opposite sides of it.

3. Draw the image of each shape. Then write the coordinates of its corners.

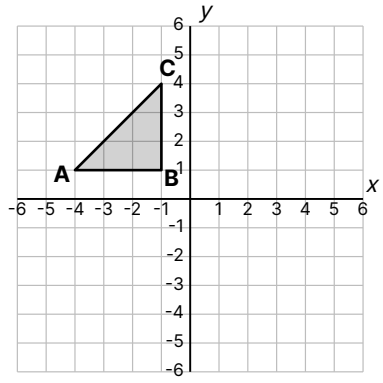
a) Reflect the shape in the y-axis.



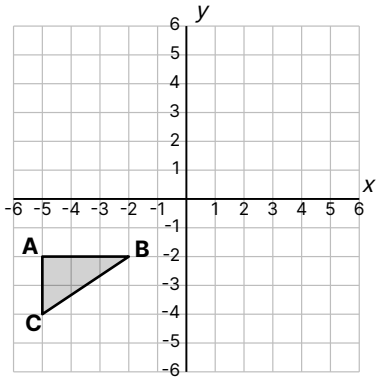
b) Reflect the shape in the x-axis.



c) Reflect the shape in the y-axis.



d) Reflect the shape in the x-axis.



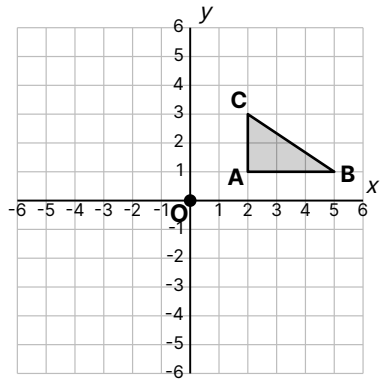
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C Rotation ●●●

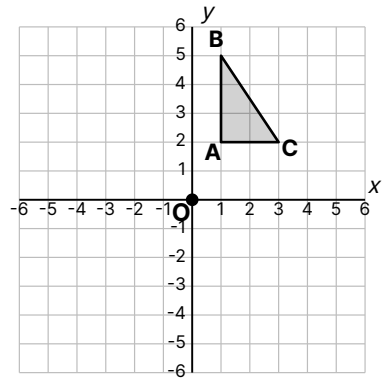
Remember: Every point turns through the same angle about the centre. Tracing paper helps: trace the shape, hold the centre still and turn.

4. Draw the image of each shape. Then write the coordinates of its corners.

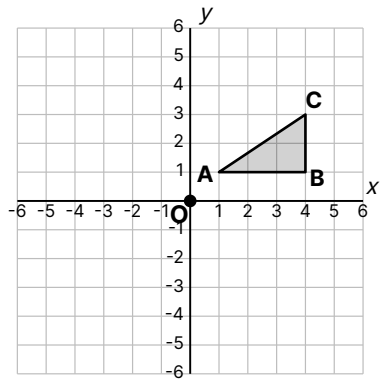
a) Rotate the shape 180° about the point O.



b) Rotate the shape 90° clockwise about the point O.

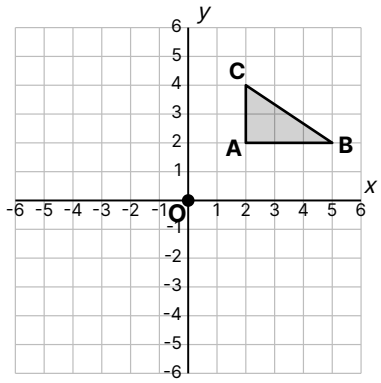


c) Rotate the shape 90° anticlockwise about the point O.



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d) Rotate the shape 90° clockwise about the point O.

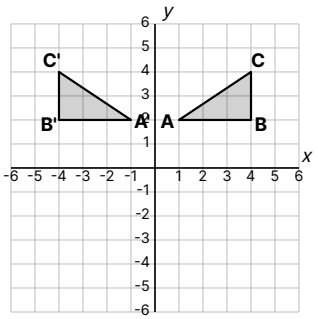


D Naming the transformation ●●●

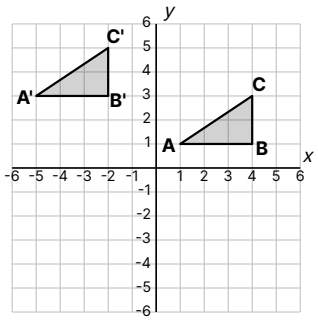
Remember: Translation: a slide. Reflection: a flip. Rotation: a turn.

5. Shape ABC has been transformed to A′B′C′. Name and describe each transformation.

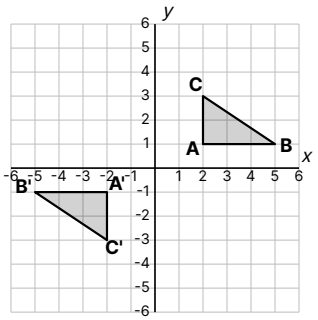
a)



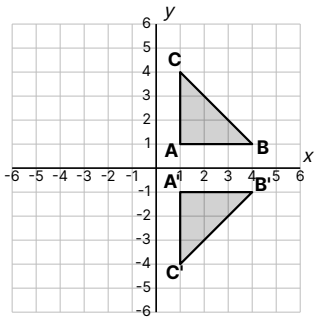
b)



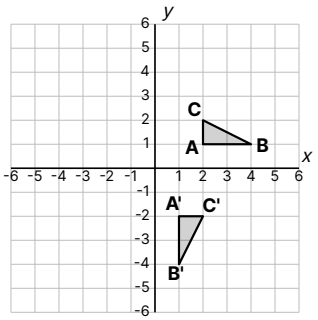
c)



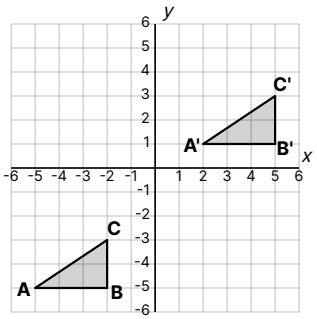
d)



e)



f)



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Extension

6. Without drawing, write the coordinates of the image of the point $P(3,\ 5)$ after each transformation.

a) Translation 6 left and 1 up _____

b) Reflection in the x-axis _____

c) Reflection in the y-axis _____

d) Rotation of 180° about (0, 0) _____

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